

Workload Estimation in Physical Human–Robot Interaction Using Physiological Measurements

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This paper uses physiological measurements to estimate human workload and effort in physical human–robot interaction. Ten subjects performed 19 consecutive task periods using the ARMin robot while difficulty was varied along two scales. Three physiological modalities were measured: electroencephalography, autonomic nervous system (ANS) responses (electrocardiography, skin conductance, respiration, skin temperature) and eye tracking. After each task period, reference workload and effort values were collected using the NASA Task Load Index. Machine learning was used to estimate workload and effort from physiological data. All three physiological modalities performed significantly better than random, particularly using nonlinear estimation algorithms. The most important ANS responses were respiration and skin conductance, while the most important electroencephalographic information was obtained from frontal and central sites. However, all three physiological modalities were outperformed by task performance and movement data. This suggests that future studies should try to demonstrate advantages of physiological measurements over other information sources.

RESEARCH HIGHLIGHTS

- Workload and effort estimation in physical human–robot interaction using physiological measurements.
- Electroencephalography, autonomic nervous system, eye tracking.
- Machine learning algorithms: linear regression, random forests.
- Good estimation accuracy, most relevant signals identified.
- Physiological measurements outperformed by task performance and movement data.

Keywords: human-robot interaction; assistive technologies; tangible interaction; virtual reality; novel interaction paradigms

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1. INTRODUCTION

As machines become more advanced, they need new ways of communicating with humans. The paradigm of affective computing posits that one important communication channel is for computers to recognize human emotions (Picard *et al.*, 2001). Such ‘affective’ machines could better interact with humans in real time. In their seminal work on the topic, Picard *et al.* (2001) demonstrated that physiological measurements such as electrocardiography can be used to differentiate between emotions. Since then, these measurements have been promoted

as an excellent emotion recognition tool, as they capture both conscious and subconscious facets of user state without the need for direct user input (Fairclough, 2009). For example, blood pressure or skin conductance may increase when the user is excited, frustrated or stressed.

One important psychological quantity in affective computing is subjective (felt or perceived) workload. To perform at an optimal level, a user should be provided with enough workload to avoid boredom, but not so much as to be overwhelmed. As the optimal workload is user-specific, it is frequently recognized through physiological measurements. Such recognition was first

applied in adaptive automation, where computers took over tasks from users in cases of high workload (see [Byrne and Parasuraman \(1996\)](#) for an early review). Work on adaptive automation has progressed considerably since then ([Ting et al., 2010](#); [Wilson and Russell, 2007](#)), and physiology-based workload recognition has expanded into other fields. For instance, games have been proposed where the computer adapts game difficulty based on the player's enjoyment, stress or workload ([Chanel et al., 2011](#); [Liu et al., 2009](#)).

Human–robot interaction was another early application of affective computing, with physiological measurements used to recognize stress, anxiety and enjoyment ([Liu et al., 2008](#); [Rani et al., 2002, 2004](#)). Recently, several physiological studies have been performed in rehabilitation robotics, with the goal of using physiological measurements to adapt workload so that patients are appropriately challenged and gain optimal benefit from rehabilitation. Closed-loop examples of physiology-based workload adaptation in upper ([Novak et al., 2011a](#)) and lower ([Koenig et al., 2011](#)) extremity rehabilitation robotics were first presented in 2011. Workload was recognized through four autonomic nervous system (ANS) responses: electrocardiography (ECG), skin conductance, respiration and skin temperature. However, these responses are very susceptible to both mental and physical workload. In physical human–robot interaction, effects of physical activity on ANS responses can overshadow indicators of mental workload ([Novak et al., 2011b](#)). For this reason, previous studies ([Novak et al., 2011a](#)) suggested combining ANS responses with measures such as eye tracking in order to improve accuracy.

Several authors have proposed alternative setups for physiology-based workload adaptation in rehabilitation, most notably [Badesa et al. \(2012\)](#) and [Guerrero et al. \(2013\)](#). While [Novak et al. \(2011a\)](#) and [Koenig et al. \(2011\)](#) use a classifier to assign discrete workload levels (low/high) to physiological responses, [Badesa et al. \(2012\)](#) and [Guerrero et al. \(2013\)](#) use fuzzy logic to estimate continuous workload values. For instance, workload could be any number between 0 and 100. Such estimation of continuous values had been used with physiological measurements in computer games ([Mandryk and Atkins, 2007](#)) and adaptive automation ([Ting et al., 2010](#)), but is not used in affective computing nearly as frequently as classification ([Novak et al., 2012](#)).

Estimation has potential advantages over classification for workload recognition in rehabilitation robotics. A classifier treats classes as unrelated, which is false when distinguishing workload levels. For example, [Koenig et al. \(2011\)](#) distinguished four workload levels (very high, high, low and very low), but the classifier was not aware that all four are levels of a single psychological quantity. [Novak et al. \(2011a\)](#) distinguished two difficulty levels (low/high), and highlighted the problem of cases near the border between low and high difficulty. This could be addressed with an estimator that would model the relationship between physiological responses and workload as a continuous function. Such continuous functions

are also intuitive for rehabilitation robots, which generally do not choose between discrete actions, but adjust their parameters between a minimum and maximum. For example, [Guerrero et al. \(2013\)](#) map the level of haptic assistance (which can be anywhere between 0 and 100%) to workload (which has a continuous range of 0 to 1). The rehabilitation game of [Novak et al. \(2011a\)](#) changes difficulty by changing the speed of a virtual ball, another variable that could be set to any value over a continuous range.

While we believe that estimation rather than classification of workload using physiological measurements is a promising approach, there are some weaknesses with the work of [Badesa et al. \(2012\)](#) and [Guerrero et al. \(2013\)](#). The fuzzy estimator of [Badesa et al. \(2012\)](#) was not validated; although its output changes during a task, the changes do not affect the task and there is no indication of whether they correspond to actual workload changes. The estimator of [Guerrero et al. \(2013\)](#), on the other hand, adapted robotic assistance and resulted in increased user enjoyment. However, there was no evaluation of whether the system intelligently estimates workload or simply increases enjoyment by providing any assistance at all: the only evaluated conditions were ‘no assistance’ and ‘physiology-triggered assistance’.

This problem was partially addressed in physical human–robot interaction by [George et al. \(2012\)](#), who classified electroencephalography (EEG) into ‘high workload’ and ‘low workload’, then activated assistance during high workload. They tested three conditions: ‘no assistance’, ‘always assistance’ and ‘physiology-based assistance’ and found physiology-triggered assistance to be intelligent, with participants reporting a high correlation between computed and felt mental workload. The authors suggested that a combination of EEG and ANS responses may allow better workload recognition.

To summarize, estimation of continuous values may be an efficient (though unproven) alternative to classification, and combining ANS responses with other physiological measurements may allow the most accurate workload estimation. Our study evaluates these possibilities by estimating workload and effort using three physiological modalities: ANS responses (ECG, skin conductance, respiration and skin temperature), EEG and eye tracking. The goal is not to provide closed-loop control of the task; it is to evaluate the accuracy of the three modalities with reference to a self-report questionnaire. This is a significant challenge not only in human–robot interaction, but in general affective computing ([Novak et al., 2012](#)). As every person reacts differently to workload, we analyze not only workload itself, but also effort put into the task, which represents the subjective reaction to workload.

Physiology-based workload estimation has another significant challenge. Most laboratory studies estimate workload in well-controlled conditions; for instance, by providing a brief rest after every task condition during which no stimuli are presented, allowing physiological effects of the task to be ‘washed out’ before the next condition. They often also induce a single

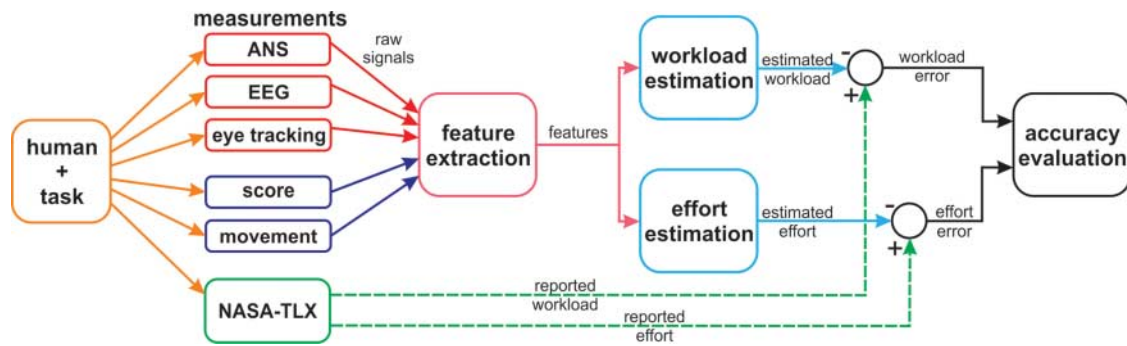


Figure 1. Study flowchart. ANS, autonomic nervous system; EEG, electroencephalography; NASA-TLX, Task Load Index questionnaire.

workload type (e.g. mental workload) while minimizing, e.g. physical activity. Such experiments are not ecologically valid, as a real session with a rehabilitation robot would not include regular rest periods and since a rehabilitation task requires both mental and physical workload. We instead designed our study to include different workload types and a single initial baseline period. All work is done with a rehabilitation robot and healthy subjects.

2. MATERIALS AND METHODS

The general flowchart of our study is shown in Fig. 1. Subjects (Section 2.1) used a robot (Section 2.2) to perform a task (Section 2.3). The NASA-TLX questionnaire (2.4) was used to collect reference workload and effort data following the study protocol (Section 2.5). During the protocol, physiological data were also recorded and features were extracted from them (Section 2.6). These features were input into machine learning algorithms that estimated the subject's workload and effort (Section 2.7). Estimated values were compared with self-reported values, and estimation accuracy was evaluated using cross-validation (Section 2.8).

2.1. Subjects

Ten healthy right-handed subjects (8 males, 2 females, 27.6 ± 3.7 years old) participated in the study. They were asked not to drink alcohol and coffee or smoke on the day of the experiment. They were also asked not to wear thick-rimmed glasses as this degraded the eye tracker performance. All gave their informed consent after the purpose and procedure of the study were explained.

2.2. Robot

We used the ARMin III (Nef *et al.*, 2009), a rehabilitation robot with an exoskeletal structure with six actuated degrees of freedom, including a hand module. The subject's dominant arm is connected to the robot with cuffs on the upper arm and

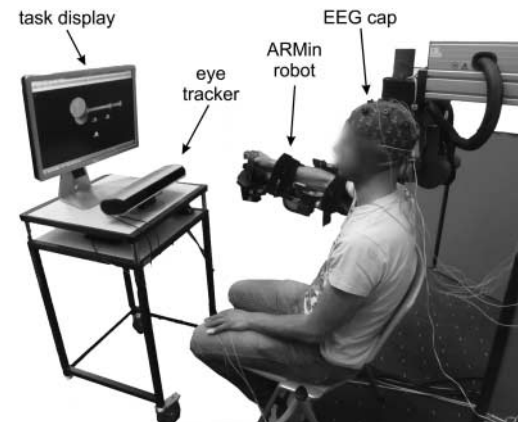


Figure 2. A subject performing the task with the ARMin robot while being monitored by physiological sensors.

forearm, and the hand is strapped to the hand module. The dimensions of the device are adjustable to each subject. Position sensors allow interaction between the subject and robot to be measured at the joint level. Gravity and friction compensation is provided by the robot, allowing the arm to be moved without resistance.

Although the ARMin III is a rehabilitation robot, we used it as a general physical human–robot interaction set-up. While rehabilitation is a relevant application for this study, we do not wish to emphasize it as the only possibility. A photo of a subject performing the task with the ARMin is shown in Fig. 2.

2.3. Task

The principle of our ‘whack-a-mole’ task is to hit monsters with a hammer before they disappear (Fig. 3). The hammer is moved around the screen with the robot's end-effector, and a ‘hitting’ movement is performed with forearm pronation. Monsters appear at one of nine locations (3×3 layout) and disappear if not hit within a certain amount of time. However, the subject should not hit all monsters: each monster has an equation attached to it, and the subject should only hit correct equations.



Figure 3. A screenshot of the task with the hammer, the monsters and the equations on the monsters.

Equations are restricted to addition and range from simple to very difficult. The probability of an equation being correct is 50%, though individual equations are generated randomly.

The task has two adjustable parameters: the equation difficulty and the frequency with which monsters are spawned. A monster can spawn every 1.5, 2.5, 4 or 6 s. Each monster remains on the screen 2.5 times the spawn interval, so there are never more than three monsters on the screen. The equation difficulty has five possible levels. Example equations are:

- (i) Level 1: $12 + 4 = 16$ (double digit + single digit, no carry).
- (ii) Level 2: $12 + 19 = 31$ (double + double digit, carry).
- (iii) Level 3: $77 + 34 = 111$ (double + double digit, carry twice).
- (iv) Level 4: $12 + 33 + 12 = 67$ (double + double + double digit, no carry).
- (v) Level 5: $45 + 33 + 63 = 141$ (double + double + double digit, carry twice).

A higher equation difficulty requires the subject to perform more complex mental arithmetic, while a higher spawn frequency gives the subject less time to move the robot and to perform the mental arithmetic. With 5 equation difficulty levels and 4 temporal difficulty levels, there are 20 possible conditions.

Initially, we also considered the possibility of force feedback, which was used in other rehabilitation robots to adapt physical workload (Novak *et al.*, 2011b). However, pretests showed that force feedback in the ARMin was either very fatiguing or had no effect on the task. Therefore, no force feedback was provided.

2.4. NASA-TLX questionnaire

To evaluate the subjects' responses to the task, we chose the NASA Task Load Index (NASA-TLX) questionnaire (Hart and

Staveland, 1988), which is extensively used in human workload studies, including physiological studies (Gonzalez *et al.*, 2012; Wilson and Russell, 2007). It consists of six scales, of which performance was omitted since objective performance measures are generally available in rehabilitation robotics. The other five scales (with simplified descriptions) are:

- (i) mental workload ('How much mental and perceptual activity was required?');
- (ii) temporal workload ('How much time pressure did you feel?');
- (iii) physical workload ('How much physical activity was required?');
- (iv) effort ('How hard did you have to work, both mentally and physically, to accomplish your level of performance?');
- (v) frustration ('How discouraged, irritated, stressed and annoyed were you?').

A computerized version of the NASA-TLX was applied several times during the task. The value on each scale was selected by moving a slider along a horizontal line marked 'very low' and 'very high' at the extremes. The slider was moved by forearm pronation/supination in the robot, and the selection was confirmed verbally. The chosen value was saved as a number between 0 and 100. After the experiment, results for each scale were linearly transformed so that the lowest reported value was transformed to 0 and the highest reported value was transformed to 100. This ensured that all subjects have the same range of reported values.

The three workload scales (mental, temporal, physical) are often correlated with each other (Hart, 2006) and difficult to separate in physical human–robot interaction (Novak *et al.*, 2011b). They were combined into *workload*—the mean of the mental, temporal and physical workload values. Though we originally planned to analyze both effort and frustration, a preliminary analysis showed that frustration in the task can be expressed as a function of workload and effort, and we therefore do not report frustration results. The two remaining self-report scales were *workload* and *effort*. Both were analyzed since *workload* depends mainly on task characteristics, while *effort* represents the individual's response to the *workload*.

2.5. Study protocol

Upon the subject's arrival, the purpose and procedure of the study were explained. Subjects sat at the ARMin and placed their arm in the cuffs. The robot settings (arm length, gravity compensation, etc.) were adapted to the subject. Subjects then practiced the game at a medium difficulty level (equation level 3, spawn once per 2.5 s) until they felt comfortable and understood the task. Subjects were required to practice for a minimum of 1 min, with all but one practicing for at least 3 min. After practice, the NASA-TLX was demonstrated. The physiological sensors were applied and calibrated, and some final instructions

were given. Subjects were told to move their arm only during the experiment, to only speak when answering the NASA-TLX and to try their best even if they thought the task was impossible.

The experiment began with a 60-s baseline condition. For the baseline, subjects were told to relax, not move, keep their eyes open and remain silent. After the baseline, subjects performed 19 60-s task periods, with each task period having a different combination of equation difficulty and monster spawn frequency. Of the 20 possible combinations, only the easiest one (level 1 equations, lowest spawn frequency) was omitted as it was found to be extremely boring. The 19 combinations were presented in a random order that was different for each subject.

Each task period was followed by the NASA-TLX before the next task period. Subjects were told to answer the NASA-TLX for the preceding 60-s period. If subjects moved their head or entire body too much during the preceding period, they were reminded to move only the arm before the next period began. After the 19th task period and NASA-TLX, the experiment was concluded.

2.6. Measurements and feature extraction

Raw data were collected from four measurement modalities: ANS, EEG, eye tracking and non-physiological measurements. Several features were extracted from these data for each 60-s baseline and task period. Each feature was calculated over the entire 60-s period. The extracted features can optionally also be normalized (Section 2.6.5).

2.6.1. ANS responses

Four ANS signals were recorded: ECG, skin conductance, respiration and skin temperature. ECG was measured with three surface electrodes. One electrode was affixed 2 cm below the right clavicle between the first and second ribs, one at the fifth intercostal space on the mid-axillary line of the left side of the body, and one at the right acromion. Respiration was measured using a thermistor flow sensor underneath the nose. Skin conductance was measured using a g.GSR sensor (g.tec Medical Engineering GmbH, Austria) at the medial phalanges of the second and third fingers of the non-dominant hand. The peripheral skin temperature was measured using a g.TEMP sensor (g.tec) at the distal phalanx of the fifth finger of the non-dominant hand. All signals were sampled at 600 Hz using a g.USBamp signal amplifier (g.tec) and filtered with an analog 50-Hz notch filter. Additional analog filters were 0.5-Hz high-pass for ECG and 30-Hz low-pass for the other three signals.

Before feature extraction, signals were visually examined for motion artifacts. Artifacts were manually removed from the respiration, temperature and ECG signals. In the respiration and temperature signals, <0.5% of the data contained artifacts. In the ECG, ~5% of the task periods required manual artifact removal. In rare cases where an R-peak could not be precisely identified in the ECG, it was defined halfway between the previous and next R-peak. Artifacts in the skin conductance

signal, however, were only removed if they were obviously artifacts; in several cases, we could not determine whether events were artifacts or actual skin conductance changes, and did not remove them.

From the ECG, intervals between two normal heartbeats (NN intervals) were extracted. Then, *mean heart rate*, the *standard deviation of NN intervals (SDNN)* and the *square root of the mean squared differences of successive NN intervals (RMSSD)* were calculated as described in Task Force (1996). These metrics have been extensively used both in general psychophysiology (Kreibitz, 2010) and physical human–robot interaction (Badesa *et al.*, 2012; Guerrero *et al.*, 2013; Novak *et al.*, 2011b).

From the skin conductance signal, we first calculated the *mean skin conductance derivative*. The signal was then filtered with a third-order Butterworth high-pass filter (cut-off: 0.05 Hz), and skin conductance responses (SCRs) were detected. An SCR is a transient increase in skin conductance whose amplitude exceeds $0.05 \mu\text{S}$ and whose peak occurs <5 s after the beginning of the increase. *SCR frequency* and *mean SCR amplitude* were calculated. SCRs are a strong indicator of both mental and physical workload (Novak *et al.*, 2011b) and have been frequently used with rehabilitation robots (Badesa *et al.*, 2012; Guerrero *et al.*, 2013). The mean skin conductance derivative represents the tonic component of skin conductance, which is more strongly affected by physical than mental workload (Novak *et al.*, 2011b).

From the respiration signal, we calculated the *mean respiratory rate* and the *standard deviation of respiratory rate*. The respiratory rate increases with arousal and workload (Brookings *et al.*, 1996; Veltman and Gaillard, 1998), while respiratory rate variability generally decreases with increasing workload but increases again if the workload becomes too high (Boiten, 1993; Novak *et al.*, 2011b).

From the temperature signal, we calculated the *final skin temperature* as the mean temperature during the last 5 s of each period. Additionally, the *mean derivative of skin temperature* was calculated over the entire period. Skin temperature generally decreases with increasing mental workload in physical human–robot interaction (Badesa *et al.*, 2012; Novak *et al.*, 2011b).

2.6.2. Electroencephalography

The EEG was measured with the g.GAMMAcap (g.tec) and g.Butterfly active electrodes. Electrodes were placed at 11 locations of the International 10–20 system: Fz, F3, F4, Cz, C3, C4, Pz, P3, P4, O1 and O2. Signals were referenced to an electrode at position FPz and grounded with an electrode on the left earlobe. These locations are commonly used in EEG studies of mental workload (Berka *et al.*, 2007; George *et al.*, 2012; Wilson and Russell, 2007). Additionally, the electro-oculogram (EOG) was recorded with two gold electrodes (Grass Technologies, USA): one to the upper right side of the right eye and one to the lower left side of the left eye. It was used only to

remove ocular artifacts in the EEG. All signals were sampled at 600 Hz using a second g.USBamp amplifier (g.tec) and filtered with an analog 50-Hz notch filter as well as an analog 1–30 Hz band-pass filter.

Ocular artifacts were removed from the EEG using an adaptive filter with EOG as the noise reference (He *et al.*, 2004). The power spectral density of each EEG channel was then calculated using Welch's method of modified periodograms (Proakis and Manolakis, 2007).

The EEG consists of several frequency bands: delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz) and gamma (30–70 Hz). EEG frequencies above 20 Hz are contaminated by electrical muscle activity (Whitham *et al.*, 2007), which is problematic for our task. In pretests, we found high beta and gamma activity in the EEG even when subjects simply moved their arm inside the robot in a circular motion. Especially, gamma power sometimes increased to over double the baseline value. We therefore did not extract features from beta and gamma bands despite their well-documented connection to mental processes.

Alpha and theta bands nonetheless contain a large amount of information about mental tasks (Klimesch, 1999). Common features for these bands are either total power in one band (Wilson and Russell, 2007) or total power in one band divided by total power in all bands (Fairclough and Venables, 2006). Since there is no consensus as to the best features, we calculated three features for each EEG channel:

- (i) alpha power divided by total power;
- (ii) theta power divided by total power;
- (iii) alpha power divided by theta power.

The total power was calculated over the 1–20 Hz band to avoid contamination by electrical muscle activity.

2.6.3. Eye tracking

Eye tracking was performed using the SMI RED 250 (SensoMotoric Instruments, Germany), a remote eye tracker placed underneath and slightly in front of the screen. This placement ensured an optimal distance between the eyes and tracker. The sampling frequency was 250 Hz. The same lighting level was maintained throughout data collection.

Feature extraction is done by the manufacturer's provided software, BeGaze 3.1, which segments signals into blinks, saccades (rapid gaze shifts from point to point) and fixations. It outputs the *number of blinks*, *number of saccades* and *number of fixations* as well as the *mean saccade duration* and *mean fixation duration*. Outside human–robot interaction, blink frequency is negatively correlated with workload (Brookings *et al.*, 1996; Van Orden *et al.*, 2001), while fixation and saccade frequencies are positively correlated with workload in visual search tasks (Van Orden *et al.*, 2001). Fixation duration changes with workload, though the type of change depends on the type of task demands (De Rivecourt *et al.*, 2008).

For saccades, BeGaze outputs the *peak saccade velocity*, a sensitive index of arousal and task difficulty (Di Stasi *et al.*, 2013). For fixations, it outputs the *mean pupil diameter* and *standard deviation of pupil diameter*. Pupil diameter increases during complex visual search tasks (Van Orden *et al.*, 2001) and with stress (Pedrotti *et al.*, 2014).

2.6.4. Non-physiological measurements

In addition to physiological features, *task score* was calculated by giving one point for each monster the subject correctly hit, subtracting one point for each monster the subject hit but should not have, and subtracting one point for each monster the subject did not hit but should have. Furthermore, *mean velocity* was calculated as the mean absolute end-effector velocity in all three directions (x, y, z) of the robot's workspace.

2.6.5. Feature normalization

Physiological features are often normalized with respect to the baseline period by subtracting the feature's baseline value from the current value and dividing the result by the baseline value (Novak *et al.*, 2012):

$$x_{\text{norm}} = \frac{x_{\text{current}} - x_{\text{baseline}}}{x_{\text{baseline}}},$$

where x can be any physiological feature.

Estimation algorithms were tested both with and without normalization, and cross-validation (Section 2.8) was used to decide whether to include normalization in the final estimation algorithm.

2.7. Estimation algorithms

For purposes of this study, we define estimation as taking extracted physiological features as inputs and outputting a continuous value (between 0 and 100) corresponding to *workload* or *effort*. The estimation is done once per task period using features calculated over the entire period. *Workload* and *effort* were estimated independently of each other, though the same procedure was used for both.

When working with ANS responses, three estimation methods are common: linear regression, fuzzy logic and neural networks (Novak *et al.*, 2012). Fuzzy logic was previously used with rehabilitation robots (Badesa *et al.*, 2012; Guerrero *et al.*, 2013), but has a major weakness: the rules underlying the system must be manually defined. As we feel that physiological responses in physical human–robot interaction are too complex to manually define accurate rules, we believe it is more appropriate to create estimation rules using supervised machine learning. Supervised machine learning is also the dominant approach in workload classification (Novak *et al.*, 2012).

We use two supervised machine learning methods for estimation: stepwise linear regression and random forests. Both require training data: examples of physiological features and their corresponding *workload* or *effort*.

2.7.1. Stepwise linear regression

Linear regression estimates *workload* or *effort* as a weighted linear sum of physiological features. The weights are calculated from training data using the least squares method. To select only the most relevant features, we used stepwise linear regression, implemented in MATLAB 2011a through the *stepwisefit* function. Stepwise linear regression uses the features' *P*-values to select the most informative features. It was set to add features until at least N_{crit} features have been included and no remaining feature has a *P*-value below p_{crit} . The parameters N_{crit} and p_{crit} are set in cross-validation (Section 2.8). Stepwise linear regression is simple and robust, but cannot handle nonlinear relationships in the data.

2.7.2. Random forest

A random forest (Breiman, 2001) is a nonlinear regression method and should have an advantage over stepwise linear regression if strong nonlinear trends are present in the data. It is an ensemble model, consisting of many individual regression trees. A single regression tree creates a branching structure of IF-THEN rules to predict the output from all possible input features, with each IF-THEN rule based on a single feature. A random forest generates N_{trees} , each trained with a randomly selected subset of N_{input} features. The individual trees' outputs are combined to generate the final output.

While individual trees are commonly used to recognize workload from physiological data, random forests and other ensemble methods are far less popular (Novak *et al.*, 2012). However, random forests are more robust with regard to overfitting than individual trees (Breiman, 2001). Random forests were implemented using the MATLAB algorithms available from <http://code.google.com/p/randomforest-matlab/> (downloaded February 2013); N_{trees} and N_{input} were set in cross-validation (Section 2.8).

2.8. Cross-validation

2.8.1. Estimation accuracy

Estimation accuracy was evaluated through cross-validation: the algorithms were trained on a subset of the available data, and then tested on a different subset of the data, ensuring generalizability to new data. We used leave-condition-out and leave-subject-out cross-validation:

- (i) Leave-condition-out cross-validation is performed separately for each subject. The estimation algorithm is trained on 18 task periods, and then tested on the remaining task period. The process is performed 19 times per subject, with each task period used as the 'test' condition once.
- (ii) Leave-subject-out cross-validation combines data from different subjects. The estimation algorithm is trained on all task periods from 9 subjects (total $19 \times 9 = 171$), and then tested on the 19 task periods

of the remaining subject. The process is performed 10 times, with each subject used as the 'test' subject once.

In both cross-validation types, the algorithm produced *workload* and *effort* estimates for all 190 task conditions. We must then determine the accuracy of these estimates. For this, we can use the self-reported *workload* and *effort* values as a reference that is assumed to be correct. This assumption is not always true, but questionnaires are, nonetheless, a common reference measure in physiological inference of psychological states (see Novak *et al.* (2012) for a review).

Estimated *workload* and *effort* were compared with self-reported values. The final result was the mean error, defined as the mean absolute difference between estimated and reported values across all 190 conditions. It was obtained separately for *workload* and *effort*. Furthermore, it was calculated separately for:

- (i) leave-condition-out and leave-subject-out cross-validation;
- (ii) each measurement modality (ANS, EEG, eye tracking, non-physiological), all three physiological modalities together and all four modalities together;
- (iii) both estimation algorithms (stepwise linear regression, random forest).

2.8.2. Setting optimal parameters

The estimation algorithms have several parameters that can be set:

- (i) Normalize features: yes/no.
- (ii) N_{crit} : 1, 3, 5, 8 (linear regression).
- (iii) p_{crit} : 0.05, 0.1 (linear regression).
- (iv) N_{trees} : 5, 10, 25, 50, 100 (random forests).
- (v) N_{input} : 2, 5, 10, 20 (random forests).

These settings must be chosen intelligently, but we again should not train and test on the same data. We therefore use a double cross-validation procedure. The data are first split into 'external' training and test groups as described in Section 2.8.1. Each external training group is further split into 'internal' training and test subgroups according to the leave-one-out principle. All possible parameter settings are evaluated in internal cross-validation, and the best settings (those with the lowest mean error) are used in external cross-validation.

As an example, let us look at leave-subject-out cross-validation:

- (i) Split subjects into nine external training and one external test subject. Perform internal cross-validation on nine training subjects.
- (ii) Internal cross-validation: eight training, one test subject. Train estimators with all possible settings, and calculate differences between reported and estimated values. Repeat over all nine subjects, with each subject used as the internal test subject once.

- (iii) Find ‘best’ settings with lowest mean error over all nine internal subjects.
- (iv) External cross-validation: Train estimator with ‘best’ settings on nine external training subjects and test on remaining subject.
- (v) Repeat over all 10 subjects, with each subject used as the external test subject once.
- (vi) Final result: mean error over all 10 subjects in external cross-validation.

2.8.3. Comparison with random inputs

Having calculated the mean errors, we can state whether a particular measurement modality is more accurate than another. However, we do not know what the ‘worst-case’ result would be. For instance, while EEG may be better than eye tracking, both may be only slightly better than random guessing. We therefore also performed cross-validation using randomly generated data as input features. The randomly generated features were not normalized, though different algorithm parameters (Section 2.8.2) were tested.

Twelve features were randomly generated for each task period: three from a normal distribution with a mean of 0 and standard deviation of 1, three from a normal distribution with a mean of 10 and standard deviation of 5, three from a uniform distribution between -1 and 1 , and three from a uniform distribution between 5 and 15 . Randomly generated features had different distributions since different physiological features also have different distributions. The specific distribution selection was entirely arbitrary.

2.8.4. Most informative features

After building our estimators, it is useful to know which features provide the most information about *workload* or *effort*. To do this, we performed stepwise linear regression ($p_{\text{crit}} = 0.05$) on data from all task periods of all 10 subjects. Unlike the regression models trained for accuracy evaluation, which only use part of the data for training, this model is trained on all available data and therefore represents the ‘optimal’ linear leave-subject-out estimator. The feature set selected by this model was considered to be the most informative.

Stepwise linear regression was performed separately for *workload* and *effort*, and for each measurement modality (ANS responses, EEG, eye tracking and non-physiological data). The resulting optimal feature set does not necessarily consist of the ‘best’ individual features; due to the principle of stepwise linear regression, it consists of features that usefully complement each other and is biased toward features with a linear relationship to *workload* or *effort*.

2.8.5. Statistical methods

We first checked whether task difficulty settings were correlated with self-report data. Pearson’s correlation coefficients between task difficulty and self-reported workloads were computed for each individual subject.

Statistical differences between mean errors were evaluated for *workload* and *effort* separately, as well as separately for leave-condition-out and leave-subject-out cross-validation. In each of these six cases (3 scales, 2 cross-validations), the mean error was calculated for each subject separately using each input modality and estimation method (linear regression, random forest). Normality was tested using the Shapiro–Wilk test. If the normality requirements were met, a two-way (modality $\times$ estimator) repeated-measures analysis of variance was performed with $N = 10$ subjects, and the Sidak correction for multiple comparisons was used for post hoc tests. The Huynh–Feldt correction was used for violations of sphericity. If normality requirements were not met, the Friedman two-way analysis of variance was used instead.

3. RESULTS

3.1. Induced workload

The mean correlation coefficient between equation difficulty and mental workload was 0.65 over all subjects (minimum 0.49, maximum 0.81). P -values of the correlation were below 0.05 for all subjects (mean 0.007, maximum 0.031). The mean correlation coefficient between monster spawn frequency and temporal workload was 0.67 over all subjects (minimum 0.49, maximum 0.82). P -values of the correlation were below 0.05 for all subjects (mean 0.009, maximum 0.032).

Before transforming workload and effort to 0–100 for each subject, the range of reported *workload* was 10.8–81.5 (mean 48.3) and the range of reported *effort* was 25.0–90.0 (mean 65.1). These values were obtained by taking the minimums, means and maximums for each subject, and then averaging them.

3.2. Estimation results

Estimation results are shown in Figs 4 and 5 for *workload* and *effort*. Results are given as mean estimation error $\pm$ standard deviation of estimation error over 10 subjects ($N = 10$). The error is always a positive number since it is the absolute difference between estimated and reported values.

For *workload* estimation (Fig. 4), non-physiological data were more accurate than any physiological modality ($P < 0.05$ in all cases). In leave-subject-out cross-validation, linear regression outperformed random forests in the case of non-physiological data ($P < 0.001$), while random forests were better for EEG ($P = 0.018$).

For *effort* estimation (Fig. 5), non-physiological data were not significantly more or less accurate than any physiological modality. There was an effect of estimation model in leave-subject-out cross-validation: estimation with random forests was significantly better than with linear regression for all inputs other than non-physiological data ($P < 0.05$ in all cases).

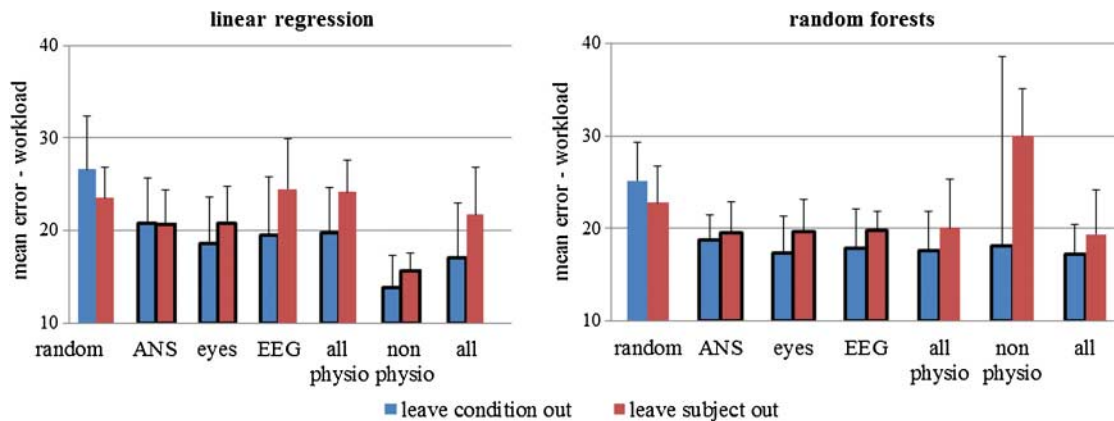


Figure 4. Mean error for *workload* estimation using different input features and linear regression (left) or random forests (right). Error bars indicate standard deviation across 10 subjects. Columns with black outlines indicate significant difference from estimation with random input data.

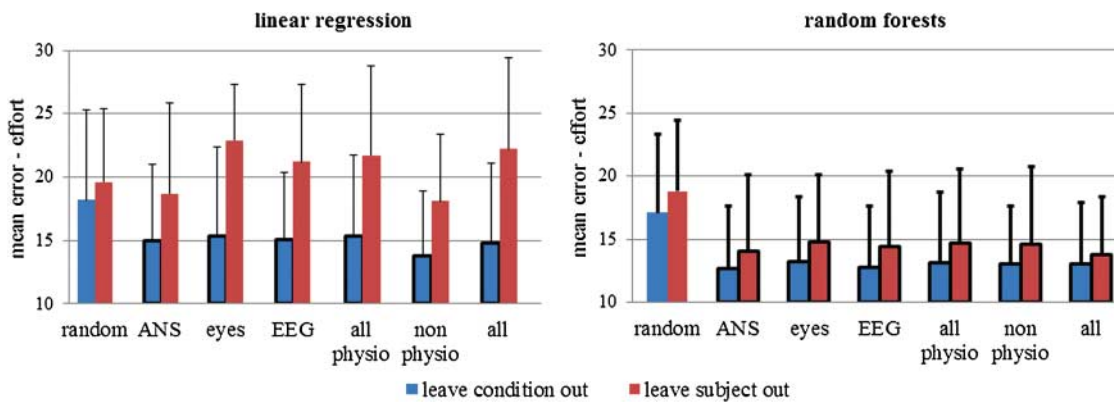


Figure 5. Mean error for *effort* estimation using different input features and linear regression (left) or random forests (right). Error bars indicate standard deviation across 10 subjects. Columns with black outlines indicate significant difference from estimation with random input data.

Combining all three physiological modalities was never significantly more accurate than the best individual physiological modality, and combining all three physiological modalities with non-physiological data did not result in a significantly lower mean error than using only non-physiological data.

3.3. Most informative features

Tables 1 and 2 show features selected by stepwise linear regression for estimation of *workload* and *effort* based on data from all 10 subjects. They are ranked from most to least informative in the final model for that measurement modality.

4. DISCUSSION

4.1. Accuracy of physiological measurements

In leave-condition-out cross-validation, all three physiological modalities estimate *workload* and *effort* significantly better than random. Random forests produce significantly smaller

errors than regression, particularly for *effort* estimation. This suggests that relationships between physiological features and workload/effort are nonlinear and that psychophysiological studies can achieve better results by taking this nonlinearity into account.

The mean error achieved for *effort* estimation is smaller than the mean error achieved for *workload* estimation. This could be due to how the two are defined: *workload* in the NASA-TLX is a description of the task characteristics (e.g. ‘how physically demanding was the task?’) while *effort* represents the individual response to the task (e.g. ‘how hard did you have to work to accomplish your level of performance?’) and should therefore be better reflected in physiological responses. However, the mean error achieved using random data is also lower for *effort*, so this difference might not be meaningful.

When using random forests, there is no significant difference between the three physiological modalities for either *workload* or *effort*. This may be due to the small sample size ($N = 10$), but it is possible that the three modalities provide similar information about the underlying psychophysiological process,

Table 1. Optimal feature set for *workload* estimation, obtained using stepwise linear regression over all 10 subjects.

	Numbers	Feature	Positive/negative	<i>P</i>
ANS	1	Mean respiratory rate	Positive	<0.001
	2	SD of respiratory rate	Positive	0.001
Eye tracking	1	Mean pupil diameter	Positive	<0.001
	2	SD of pupil diameter	Positive	<0.001
EEG	1	F4 alpha/total power	Negative	<0.001
	2	C4 alpha/total power	Positive	<0.001
	3	C3 alpha/total power	Negative	0.014
Non-physiological data	1	Score	Negative	<0.001
	2	Up/down movement	Positive	<0.001

For each feature, we list whether it is positively or negatively correlated with *workload* and its *p*-value in the final model.

Table 2. Optimal feature set for *effort* estimation, obtained using stepwise linear regression over all 10 subjects.

	Numbers	Feature	Positive/negative	<i>P</i>
ANS	1	SCR frequency	Positive	<0.001
	2	Mean respiratory rate	Positive	<0.001
	3	Mean SCR amplitude	Positive	0.047
Eye tracking	1	Number of fixations	Negative	<0.001
	2	Mean fixation duration	Positive	<0.001
	3	Number of blinks	Negative	<0.001
EEG	1	Cz alpha/theta power	Positive	<0.001
	2	O2 theta/total power	Positive	<0.001
	3	O2 alpha/total power	Positive	<0.001
	4	P3 theta/total power	Negative	<0.001
	5	O2 alpha/theta power	Negative	<0.001
	6	P3 alpha/total power	Negative	<0.001
	7	Fz theta/total power	Negative	0.0029
	8	C3 alpha/total power	Positive	0.028
Non-physiological data	1	Up/down movement	Positive	<0.001
	2	Left/right movement	Negative	<0.001
	3	Score	Negative	0.0064
	4	Forward/backward movement	Positive	0.046

For each feature, we list whether it is positively or negatively correlated with *effort* and its *p*-value in the final model.

making them redundant. Furthermore, combining the three modalities did not increase estimation accuracy over using a single modality. This may be due to redundancy, but may also be due to the large number of input features making it difficult to find the optimal estimation rules. The results, at least, do not support the hypothesis that combining physiological modalities can improve estimation accuracy.

4.2. Most informative physiological features

Among ANS responses, respiration and skin conductance are the most informative signals. Respiration is connected to *workload* and *effort*, while skin conductance is related to *effort*. This partially agrees with previous studies in robotics: Novak *et al.* (2011b), Badesa *et al.* (2012) and Guerrero *et al.* (2013)

emphasized skin conductance as a sensitive indicator of both physical and mental demands. However, while Novak *et al.* (2011b) also obtained good results for respiration, Badesa *et al.* (2012) did not. This may be because both this study and the Novak *et al.* (2011b) study involve more variations in physical workload than the work of Badesa *et al.* (2012), and Novak *et al.* (2011b) showed that respiration is relatively robust with regard to varying physical workload.

For eye tracking, pupil diameter is the most sensitive to *workload*, while blink and fixation frequencies are the most sensitive to *effort*. However, a follow-up analysis of *workload* showed that in the first step of the feature selection, the number of fixations was only slightly less informative than pupil diameter. Once pupil diameter was included, the number of fixations contributed little additional information and was

considered redundant. A similar result was found in a follow-up analysis of *effort*, leading us to believe that pupil diameter, blink frequency and fixation frequency provide very similar, redundant information in our task.

For EEG, *workload* information is mostly found in central and frontal areas, while *effort* information is found at many different sites. In a follow-up analysis, we found that excluding occipital and parietal signals had little effect on *effort* estimation results, but that excluding frontal or central signals noticeably decreased the accuracy. Therefore, we conclude that both frontal and central sites should be used in physical human–robot interaction.

4.3. Accuracy of non-physiological measurements

Task score and movement velocity estimated *workload* and *effort* better than any physiological modality. Furthermore, combining non-physiological and physiological data did not give a lower estimation error than non-physiological data alone. This implication is somewhat troublesome for physiology-based workload estimation in rehabilitation. If physiological measurements do not give more information than score and movement velocity, which are readily available in rehabilitation robotics, there is no point in using them, especially since performance-based difficulty adaptation has been successfully demonstrated in rehabilitation (Cameirão *et al.*, 2010).

Interestingly, our current results are worse than previous classification work in rehabilitation robotics. In Novak *et al.* (2011a), physiological measurements usefully supplemented task performance information. In Koenig *et al.* (2011), physiological measurements outperformed task performance in several cases. Studies outside robotics (e.g. Hussain *et al.*, 2014) have also found that task performance can outperform physiological measurements, but showed an improvement when combining physiological and non-physiological measures.

A possible reason for the worse result (physiological measurements are less accurate than task score) in this study is the greater challenge faced: there are 19 consecutive task periods compared with, e.g. 6 in Novak *et al.* (2011a), and only a single initial baseline period. As some physiological responses (e.g. skin conductance) exhibit noticeable drift, laboratory studies commonly include frequent rest conditions where physiological responses can return to a baseline state. This is not possible in an actual rehabilitation session, where the goal is to maximize exercise intensity. Any additional baseline periods would not be accepted by therapists since they would directly decrease exercise intensity. This study design, however, likely decreased accuracy of physiological measurements since finding an estimation rule that holds true across 19 consecutive conditions can be very difficult. In a real rehabilitation session, it may be possible to use parts of the session (e.g. when the patient rests and talks to the therapist) as baseline periods.

A second reason for the poor result may be the workload levels used in the study. When workload increases from low to

moderate levels, task performance often remains stable while perceived workload increases—performance and workload dissociate (Vidulich and Tsang, 2012). On the other hand, when workload changes from moderate to high, physiological measures and task performance often correlate. It is possible that the task used in our study was generally at least moderately demanding, so changes in performance were as predictive of workload as physiological measurements (or more so). This is a weakness of our task, but also presents a general problem for physiological measurements in rehabilitation: as patients should be exercising at high intensities, physiological measurements and task performance can be expected to correlate.

In general, rehabilitation robotics represents a difficult application for psychophysiological measurements. Psychophysiological assessment may be more suitable for applications where physical workload is low and task performance information is unavailable or irrelevant. This is rarely the case in rehabilitation, where excess workload would result in degraded performance.

4.4. How accurate is enough?

A common problem in applied psychophysiological studies is that relationships between inference accuracy and user acceptance are unclear, as noted in a recent review (Novak *et al.*, 2012). When classifying psychological states, it is hard to say whether, for example, 80% classification accuracy is sufficient in a two-class (low/high workload) scenario without implementing the classifier online and examining user acceptance.

Determining the acceptable accuracy is an even bigger problem when using estimation rather than classification, as accuracy metrics for estimation are underdeveloped and unintuitive compared with metrics for classification (Novak *et al.*, 2012). For example, perfect classification accuracy is always 100%, and random classification accuracy is 100% divided by the number of classes (assuming all classes are equally probable). In our estimation study, the mean error using random data is 22.8 for *workload* and 17.1 for *effort*; so different quantities can have different ‘random’ error levels. Furthermore, our lowest mean errors are 13.9 for *workload* estimation (using non-physiological data) and 12.7 for *effort* estimation (using ANS). This represents a reduction of ~40 and 25% over random errors, respectively. However, it is difficult to even visualize what a mean error of 17 points or an error reduction of 40% represents. The lowest realistically achievable mean error is definitely not zero: that would not only require perfect knowledge of psychophysiological relationships, but would also require subjects to perfectly identify their own *workload* or *effort* via the NASA-TLX.

As estimation of continuous psychological values from physiological measurements is still relatively unexplored, our study is one of the first to raise these issues specifically with regard to estimation, and it is difficult to say whether our results are ‘good’ or ‘bad’ in an absolute sense. There are,

however, some things that could be done. First, we could try to relate estimation accuracy to classification accuracy by discretizing *workload* and *effort* into low/high and using the same input features to perform classification in addition to estimation. This would allow better comparison with previous classification studies. Secondly, we could simply create the most accurate estimation algorithm possible, and then test user satisfaction of the algorithm. While this would not tell us the minimum acceptable accuracy, it would allow us to say whether a particular accuracy is acceptable for a given application. A step in this direction was taken by Guerrero *et al.* (2013), who showed that their rehabilitation robotics algorithm can increase enjoyment in healthy subjects.

While not as popular as classification, psychophysiological estimation has been suggested as a promising approach in rehabilitation robotics. However, proponents of estimation should consider the additional problems that it introduces with regard to accuracy evaluation.

4.5. Closing the loop

Our study focused on psychophysiological inference, and therefore studied *workload* and *effort* estimation in an open-loop setting. Ultimately we would need to close the loop, with the robot and virtual task adapting in response to the inference results. With our setup, the intuitive way to do this is to change the task difficulty, with the robot behaving the same in all cases—similarly to Novak *et al.* (2011a) and Koenig *et al.* (2011).

There is another possibility: change the level of haptic feedback, for example, George *et al.* (2012), who turned the robot's haptic assistance on and off depending on workload. In a more complex implementation, Guerrero *et al.* (2013) treated the level of haptic assistance as a continuous variable, providing different levels depending on workload. Shirzad and Van der Loos (2013), on the other hand, demonstrated the opposite approach: rather than provide assistance, the robot's haptic feedback amplifies the user's errors depending on workload. While this represents an interesting possibility for rehabilitation robotics, it could not be done based on our study, which does not provide haptic feedback.

4.6. Study limitations

Though we did our best to select relevant features for each physiological modality, other features may result in higher estimation accuracy. For example, sympathetic and parasympathetic power bands were not analyzed in the ECG since they require at least 2-min uninterrupted task periods (Task Force, 1996). However, they may provide additional information, as may e.g. more complex analysis of the respiration signals. Furthermore, the EEG was contaminated with muscle and motion artifacts that could not simply be cut from the signal due to the use of spectral features. Future studies

should consider reducing artifacts through methods such as independent component analysis.

Reference values of *workload* and *effort* were obtained through the NASA-TLX questionnaire. Though the NASA-TLX is well established, it has some weaknesses (Hart, 2006). For example, subjects likely rated a task period differently when preceded by a difficult period than when preceded by an easy one. Furthermore, since subjects practiced the task as they progressed through the task periods, a difficulty setting was likely rated as easier if it appeared toward the end of the experiment. On the other hand, increased fatigue may cause physical workload ratings to increase with time. These weaknesses should be taken into account when interpreting the results, though we feel that they are not only a weakness of our study, but a fundamental problem of applying psychophysiological measurements to motor rehabilitation. Rehabilitation is a process of motor learning, and the patient should be learning throughout a session. Furthermore, as exercise intensity would be high, fatigue effects are unavoidable. A final NASA-TLX weakness specific to our study was the high frequency of its use. Every NASA-TLX response requires a break from the task during which physiological responses may drift unpredictably. This additional drift is an unrealistic factor that would not appear in a real application, but may have contributed to the poor accuracy of physiological inference.

Though it is difficult to separate effects such as habituation, physiological drift and fatigue using the data we have collected, we can at least test for temporal trends. In a final follow-up analysis, we checked whether NASA-TLX scales and physiological features are significantly correlated with task period (1–19). Results were mixed, with large differences among subjects. For example: in two subjects, self-reported *workload* and *effort* decrease significantly with time, but in another subject, workload *increases* significantly with time. Four subjects showed physiological trends consistent with decreasing workload (decreasing heart rate, respiration rate and pupil diameter over time). In eight subjects, skin temperature increased significantly over time, which is likely due to the body heating up with sustained exercise. Several time-related factors are therefore present in the study, though they do not affect all subjects in the same way.

5. CONCLUSIONS

Our contributions to the state of the art are as follows. We used three physiological modalities (ANS, EEG and eye tracking) to estimate workload and effort in physical human–robot interaction. Previous studies in this field focused on a single physiological modality (mainly ANS) and a single psychological quantity. Psychophysiological inference was performed using estimation of continuous values rather than the more popular classification into discrete classes. We evaluated

the accuracy of such physiological estimation, comparing the three modalities to each other and to non-physiological data sources. Finally, we used both linear and nonlinear estimation methods, and identified the most relevant signals for each modality.

All three physiological modalities estimated workload and effort significantly better than random. Since our study was conducted in realistic, relatively uncontrolled environmental conditions, this result is encouraging. Among ANS responses, respiration and skin conductance appear to be the most relevant for physical human–robot interaction. For EEG, measurements should be taken from frontal and central sites. Furthermore, nonlinear estimation methods such as random forests should be used to obtain the best results.

No physiological modality was significantly more accurate than another, and combining multiple physiological modalities did not increase accuracy. Our findings thus do not support suggestions that combining physiological modalities may be beneficial. Furthermore, non-physiological data (task score and movement velocity) give a more accurate estimate of workload and effort than any physiological modality, and combining non-physiological data with physiological data does not improve accuracy further. Therefore, future studies in robotics should not only aim to show that psychophysiological inference is possible. Instead, they should also try to demonstrate concrete advantages of physiological measurements over other, more easily obtained data such as task performance.

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